CS229
Problem Set 1

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Problem 1

(a) The average empirical loss for logistic regression:

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log\left(h_\theta(x^{(i)})\right) + (1-y^{(i)})\log\left(1-h_\theta(x^{(i)})\right)$$

where $y^{(i)} \in \{0,1\}$, $h_\theta(x) = g(\theta^T x)$ and $g(z) = 1/(1+e^{-z})$. As $g'(z) = g(z)(1-g(z))$, therefore the gradient is

$$\begin{aligned}
\nabla_\theta J(\theta) &= -\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \frac{h_\theta(x^{(i)})(1-h_\theta(x^{(i)}))x^{(i)}}{h_\theta(x^{(i)})} + (1-y^{(i)})\frac{-h_\theta(x^{(i)})(1-h_\theta(x^{(i)}))x^{(i)}}{\left(1-h_\theta(x^{(i)})\right)} \\
&= -\frac{1}{m}\sum_{i=1}^{m} y^{(i)}(1-h_\theta(x^{(i)}))x^{(i)} - (1-y^{(i)})h_\theta(x^{(i)})x^{(i)} \\
&= -\frac{1}{m}\sum_{i=1}^{m}\left(y^{(i)} - h_\theta(x^{(i)})\right)x^{(i)}
\end{aligned}$$

$$\begin{aligned}
H &= -\frac{1}{m}\nabla_\theta \sum_{i=1}^{m}\left(y^{(i)} - h_\theta(x^{(i)})\right)x^{(i)} \\
&= \frac{1}{m}\sum_{i=1}^{m}\nabla_\theta\left(h_\theta(x^{(i)})x^{(i)}\right) \\
&= \frac{1}{m}\sum_{i=1}^{m}(1-h_\theta(x^{(i)}))(h_\theta(x^{(i)})x^{(i)}(x^{(i)})^T.
\end{aligned}$$

For any vector $z \in \mathbb{R}^n$, we want to prove

$$z^T H z \geq 0.$$

First take

$$z^T H = \begin{bmatrix}\sum_{i=1}^n z_i H_{i1} & \cdots & \sum_{i=1}^n z_n H_{in}\end{bmatrix}$$

Thus,

$$\begin{aligned}
z^T H z &= \sum_{j=1}^{n} z_j \sum_{i=1}^{n} z_i H_{ij} \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j H_{ij} \\
&= \frac{1}{m}\sum_{k=1}^{m} h_\theta(x^{(k)})(1-h_\theta(x^{(k)}))\sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j x_i^{(k)} x_j^{(k)}
\end{aligned}$$

Note that

$$\sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j x_i^{(k)} x^{(k)} = \left(\sum_{i=1}^{n} z_i x_i^{(k)}\right)\cdot\left(\sum_{j=1}^{n} z_j x_j^{(k)}\right) = (z^T x^{(k)})^2 \geq 0$$

And note that as $h_\theta \in [0,1]$, $h_\theta(x^{(k)})(1-h_\theta x^{(k)})$ is also greater or equal to zero. Hence, $z^T H z \geq 0$.

(b) See code.

(c)

Problem 1 continued on next page. . .

$$
\begin{aligned}
p(y=1|x;\phi,\mu_0,\mu_1,\Sigma) &= \frac{p(x|y=1)p(y=1)}{p(x)} \\
&= \frac{p(x|y=1)p(y=1)}{p(x|y=0)p(y=0)+p(x|y=1)p(y=1)} \\
&= 1/\left(1+\frac{p(x|y=0)p(y=0)}{p(x|y=1)p(y=1)}\right) \\
&= 1/\left(1+\exp\left(\frac{1-\phi}{\phi}\frac{1}{2}((x-\mu_1)^T\Sigma^{-1}(x-\mu_1)-(x-\mu_0)^T\Sigma^{-1}(x-\mu_0))\right)\right) \\
&= 1/\left(1+\exp\left((\mu_0-\mu_1)^T\Sigma^{-1}x+\frac{1}{2}(\mu_1^T\Sigma^{-1}\mu_1-\mu_0^T\Sigma^{-1}\mu_0)+\log\frac{1-\phi}{\phi}\right)\right) \\
&= \frac{1}{1+\exp\left(-(\theta^T x+\theta_0)\right)},
\end{aligned}
$$

where $\theta^T = (\mu_1-\mu_0)^T\Sigma^{-1}$ and $\theta_0 = -\frac{1}{2}(\mu_1^T\Sigma^{-1}\mu_1-\mu_0^T\Sigma^{-1}\mu_0)-\log\frac{1-\phi}{\phi}$.

(d)

$$
\begin{aligned}
\ell(\phi,\mu_0,\mu_1,\Sigma) &= \sum_{i=1}^m \log\left(p(x^{(i)}|y^{(i)};\mu_0,\mu_1,\Sigma)\right)+\log\left(p(y^{(i)};\phi)\right) \\
&= \sum_{i=1}^m -\frac{1}{2}(x^{(i)}-\mu_{y^{(i)}})^T\Sigma^{-1}(x^{(i)}-\mu_{y^{(i)}})-\log((2\pi)^{n/2}|\Sigma|^{1/2})+\log(\phi^{y^{(i)}}(1-\phi)^{1-y^{(i)}})
\end{aligned}
$$

w.r.t. ϕ:

$$
\begin{aligned}
\frac{\partial}{\partial\phi}\ell(\phi,\mu_0,\mu_1,\Sigma) &= \frac{\partial}{\partial\phi}\sum_{i=1}^m \log\left(\phi^{y^{(i)}}\right)+\log\left((1-\phi)^{1-y^{(i)}}\right) \\
&= \sum_{i=1}^m \frac{y^{(i)}}{\phi}-\sum_{i=1}^m\frac{1-y^{(i)}}{1-\phi}=0 \\
\frac{1}{\phi}\sum_{i=1}^m 1\{y^{(i)}=1\} &= \frac{1}{1-\phi}\sum_{i=1}^m 1\{y^{(i)}=0\} \\
\left(\frac{1}{\phi}-1\right)\sum_{i=1}^m 1\{y^{(i)}=1\} &= m-\sum_{i=1}^m 1\{y^{(i)}=1\} \\
\left(\frac{1}{\phi}-1\right) &= m/\sum_{i=1}^m 1\{y^{(i)}=1\}-1 \\
\phi &= \frac{1}{m}\sum_{i=1}^m 1\{y^{(i)}=1\}
\end{aligned}
$$

w.r.t. μ_0

$$
\begin{aligned}
\nabla_{\mu_0}\ell(\phi,\mu_0,\mu_1,\Sigma) &= \nabla_{\mu_0}\sum_{i=1}^m -\frac{1\{y^{(i)}=0\}}{2}(x^{(i)}-\mu_0)^T\Sigma^{-1}(x^{(i)}-\mu_0) \\
&= \sum_{i=1}^m 1\{y^{(i)}=0\}(x^{(i)}-\mu_0)=0 \\
\sum_{i=1}^m 1\{y^{(i)}=0\}x^{(i)} &= \mu_0\sum_{i=1}^m 1\{y^{(i)}=0\} \\
\mu_0 &= \frac{\sum_{i=1}^m 1\{y^{(i)}=0\}x^{(i)}}{\sum_{i=1}^m 1\{y^{(i)}=0\}}
\end{aligned}
$$

Problem 1 continued on next page...

Similar for μ_1, we have

$$\mu_1 = \frac{\sum_{i=1}^m 1\{y^{(i)} = 1\}x^{(i)}}{\sum_{i=1}^m 1\{y^{(i)} = 1\}}$$

w.r.t. Σ, consider Σ as a scalar σ^2, hence:

$$\begin{aligned}\frac{\partial}{\partial\Sigma}\ell(\phi, \mu_0, \mu_1, \Sigma) &= \frac{\partial}{\partial\Sigma}\sum_{i=1}^m -\frac{1}{2}(x^{(i)} - \mu_{y^{(i)}})^T\Sigma^{-1}(x^{(i)} - \mu_{y^{(i)}}) - \frac{1}{2}\log|\Sigma| \\ &= -\sum_{i=1}^m \frac{1}{2}(x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T\Sigma^{-2} + \frac{1}{2\Sigma} = 0\end{aligned}$$

$$-\sum_{i=1}^m \frac{1}{2}(x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T + \frac{m}{2}\Sigma = 0$$

$$\Sigma = \frac{1}{m}\sum_{i=1}^m (x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T$$

(e) See code.
(f) See Figure 1.

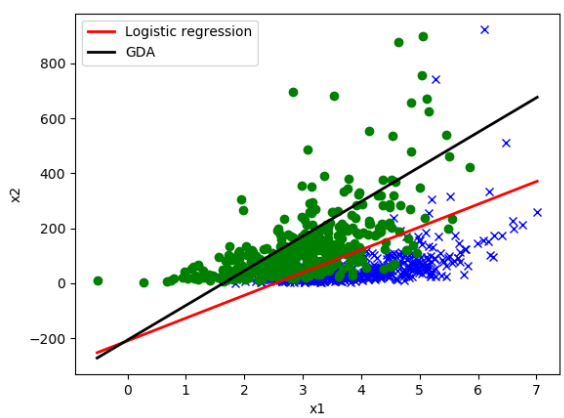

Figure 1: Dataset 1

(g) See Figure 2.
GDA is much poorer than logistic regression for dataset 1, but performs almost as well as logistic regression for dataset 2. This is because GDA assumes a Gaussian distribution of the features of the 2 classes. Thus, for dataset 1, which is not Gaussian, GDA fails.

Problem 2

(a)

$$\begin{aligned}p(y^{(i)} = 1|t^{(i)} = 1, x^{(i)}) &= p(y^{(i)} = 1|t^{(i)} = 1) \\ p(y^{(i)} = 1|x^{(i)}) &= p(y^{(i)} = 1, t^{(i)} = 1|x^{(i)}) + p(y^{(i)} = 1, t^{(i)} = 0|x^{(i)}) \\ &= p(y^{(i)} = 1, t^{(i)} = 1|x^{(i)}) \\ &= p(y^{(i)} = 1|x^{(i)}, t^{(i)} = 1)p(t^{(i)} = 1|x^{(i)}) \\ &= \alpha p(t^{(i)} = 1|x^{(i)}),\end{aligned}$$

Problem 2 continued on next page. . .

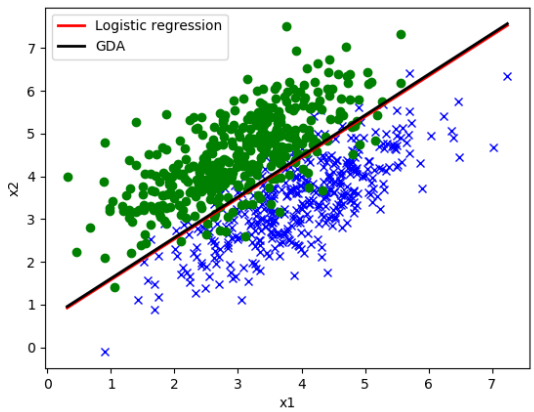

Figure 2: Dataset 2

where $\alpha = p(y^{(i)} = 1|x^{(i)}, t^{(i)} = 1) = p(y^{(i)} = 1|t^{(i)} = 1)$.
(b) For $x^{(i)} \in V_+$,

$$\begin{aligned} h(x^{(i)}) &\approx p(y^{(i)} = 1|x^{(i)}) \\ &= \alpha p(t^{(i)} = 1|x^{(i)}), \text{ use the conclusion from (a)} \\ &\approx \alpha \end{aligned}$$

(c) See code and Figure 3

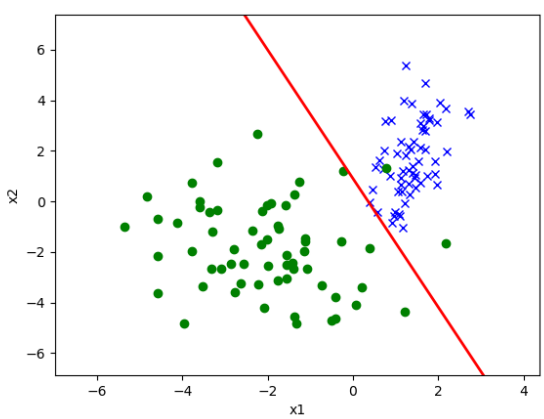

Figure 3: Decision boundary from part (c) on the test set

(d) See code and Figure 4
(e) See code and Figure 5

Problem 3

(a) Consider the Poisson distribution

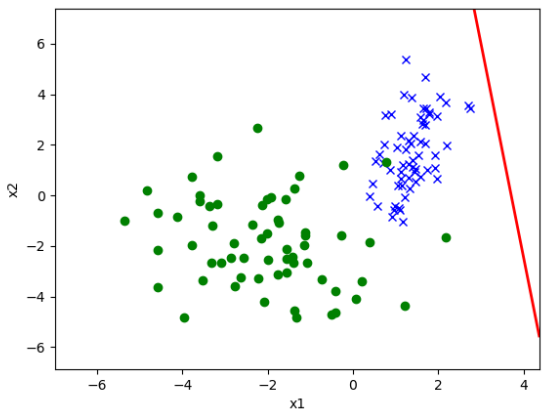

Figure 4: Decision boundary from part (d) on the test set

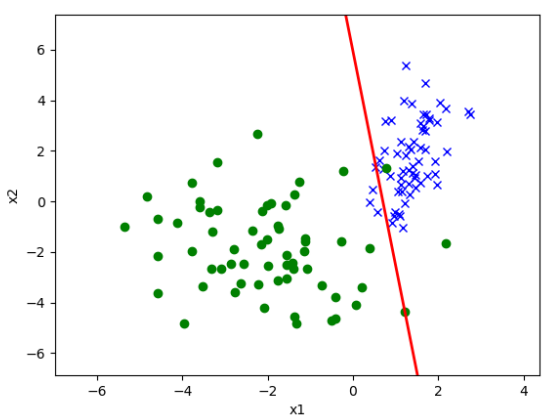

Figure 5: Decision boundary from part (e) on the test set

$$\begin{aligned} p(y;\lambda) &= \frac{e^{-\lambda}\lambda^y}{y!} \\ &= \frac{1}{y!}\exp\left(y\ln\lambda - \lambda\right) \\ &= b(y)\exp(\eta^T T(y) - a(\eta)), \end{aligned}$$

where

$$\begin{aligned} b(y) &= 1/y! \\ \eta &= \ln\lambda \\ T(y) &= y \\ a(\eta) &= e^\eta = \lambda \end{aligned}$$

(b) Hypothesis is

$$h_\theta(x) = \mathbb{E}[y|x;\theta] = \lambda = e^\eta = \exp\left(\theta^T x\right)$$

(c) The log-likelyhood of an example is

$$\begin{aligned} \ell(\theta) &= \log p(y^{(i)}|x^{(i)};\theta) \\ &= \log\left(\frac{1}{y^{(i)}!}\exp\left(y^{(i)}\ln\lambda - \lambda\right)\right) \\ &= \log\frac{1}{y^{(i)}!} + y^{(i)}\ln\lambda - \lambda \\ &= \log\frac{1}{y^{(i)}!} + \theta^T x^{(i)} y^{(i)} - \exp(\theta^T x^{(i)}) \end{aligned}$$

Its derivative w.r.t. θ_j is

$$\begin{aligned} \frac{\partial\ell(\theta)}{\partial\theta_j} &= x_j^{(i)} y^{(i)} - x_j^{(i)}\exp(\theta^T x^{(i)}) \\ &= x_j^{(i)}\left(y^{(i)} - h_\theta(x^{(i)})\right) \end{aligned}$$

Hence, the stochastic gradient ascent is

```
Loop {
    For every 1 ≤ i ≤ m {
        θ := θ + α (y^(i) − h_θ(x^(i))) x
    }
}
```

(d) See code.

Problem 4

(a)

$$\begin{aligned} \int p(y;\eta)\mathrm{d}y &= 1 \\ \frac{\partial}{\partial\eta}\int p(y;\eta)\mathrm{d}y &= 0 \end{aligned}$$

Problem 4 continued on next page. . .

$$\begin{aligned}
\frac{\partial}{\partial \eta}\int p(y;\eta)\mathrm{d}y &= \int \frac{\partial}{\partial \eta} p(y;\eta)\mathrm{d}y \\
&= \int b(y)(y-a'(\eta))\exp(\eta y - a(\eta))\mathrm{d}y \\
&= \int yb(y)\exp(\eta y - a(\eta))\mathrm{d}y - a'(\eta)\int b(y)\exp(\eta y - a(\eta))\mathrm{d}y \\
&= \mathbb{E}[y;\eta] - a'(\eta) = 0
\end{aligned}$$

Hence,

$$\mathbb{E}[Y|X;\theta] = a'(\eta)$$

(b)

$$\begin{aligned}
\frac{\partial}{\partial \eta}\mathbb{E}[Y|X;\theta] &= \frac{\partial}{\partial \eta}\int yb(y)\exp(\eta y - a(\eta))\mathrm{d}y \\
&= \int \frac{\partial}{\partial \eta} yb(y)\exp(\eta y - a(\eta))\mathrm{d}y \\
&= \int yb(y)(y-a'(\eta))\exp(\eta y - a(\eta))\mathrm{d}y \\
&= \int y^2 b(y)\exp(\eta y - a(\eta))\mathrm{d}y - a'(\eta)\int yb(y)\exp(\eta y - a(\eta))\mathrm{d}y \\
&= \mathbb{E}[Y^2|X;\theta] + \mathbb{E}[Y|X;\theta]^2 \\
&= \mathrm{Var}[Y^2|X;\theta]
\end{aligned}$$

And note that $\frac{\partial}{\partial \eta}\mathbb{E}[Y|X;\theta] = a''(\eta)$, therefore,

$$\mathrm{Var}[Y|X;\theta] = a''(\eta)$$

(c) Take one example $(x^{(k)}, y^{(k)})$ from the data set, the gradient of the loss function is

$$\begin{aligned}
\nabla_\theta \ell(\theta) &= -\nabla_\theta \log\left(b(y^{(k)})\exp(\theta^T x^{(k)} y^{(k)} - a(\theta^T x^{(k)}))\right) \\
&= -\nabla_\theta\left(\log[b(y^{(k)})] + \theta^T x^{(k)} y^{(k)} - a(\theta^T x^{(k)})\right) \\
&= -x^{(k)} y^{(k)} + x^{(k)} a'(\eta)
\end{aligned}$$

Thus, the Hessian is

$$\begin{aligned}
H &= \nabla_\theta[-x^{(k)} y^{(k)} + x^{(k)} a'(\eta)] \\
&= a''(\eta) x^{(k)} (x^{(k)})^T
\end{aligned}$$

For any vector z that has the same dimension as x,

$$\begin{aligned}
z^T H z &= \sum_i \sum_j z_i H_{ij} z_j \\
&= a''(\eta)\sum_i \sum_j z_i x_i^{(k)} x_j^{(k)} z_j \\
&= a''(\eta)\sum_i z_i x_i^{(k)} \sum_j x_j^{(k)} z_j \\
&= a''(\eta)(z^T x^{(k)})^2 \\
&= (z^T x^{(k)})^2 \mathrm{Var}[Y|X;\theta] \geq 0.
\end{aligned}$$

Note that $\ell(\theta) = \sum_k \log(-b(y^{(k)})\exp(\theta^T x^{(k)} y^{(k)} - a(\theta^T x^{(k)}))$. As each term has a positive Hessian, the sum also has a positive Hessian. That is to say, the loss function $\ell(\theta)$ is PSD.

Problem 5

(a)
i. Let the matrix $W \in \mathbb{R}^{m\times m}$ be

$$W = \text{diag}(w^{(1)}/2, w^{(2)}/2, \cdots, w^{(m)}/2),$$

therefore,

$$\begin{aligned} J(\theta) &= \sum_{i=1}^{m} (\theta^T x^{(i)} - y^{(i)}) W_{ii} (\theta^T x^{(i)} - y^{(i)}) \\ &= \sum_{i=1}^{m}\sum_{j=1}^{m} (\theta^T x^{(i)} - y^{(i)}) W_{ii} (\theta^T x^{(j)} - y^{(j)}) \\ &= (X\theta - y)^T W (X\theta - y), \end{aligned}$$

where $X \in \mathbb{R}^{m\times n}$ has entries $X_{ij} = x_j^{(i)}$.
ii.

$$\begin{aligned} \nabla_\theta J(\theta) &= \nabla_\theta (X\theta - y) \nabla_{X\theta - y}[(X\theta - y)^T W (X\theta - y)] \\ &= X^T(\nabla_{X\theta-y}(X\theta - y) W (X\theta - y) + \nabla_{X\theta-y}[W(X\theta - y)](X\theta - y)) \\ &= X^T[W(X\theta - y) + W^T(X\theta - y)] \\ &= X^T(W + W^T)(X\theta - y) \\ &= 2X^T W (X\theta - y), \text{(as W is a diag matrix)} \end{aligned}$$

Let

$$\begin{aligned} \nabla_\theta J(\theta) &= 0 \\ X^T W X \theta &= X^T W y \\ \theta &= (X^T W X)^{-1} X^T W y \end{aligned}$$

iii. The log likelihood is

$$\begin{aligned} \ell(\theta) &= \log\left(\prod_{i=1}^{m} p(y^{(i)}|x^{(i)};\theta)\right) \\ &= \sum_{i=1}^{m} -\log(\sqrt{2\pi}\sigma^{(i)}) - \frac{(y^{(i)} - \theta^T x^{(i)})^2}{2(\sigma^{(i)})^2} \end{aligned}$$

As $-\log(\sqrt{2\pi}\sigma^{(i)})$ term is constant, to maximize the log-likelihood is to minimize

$$\frac{1}{2}\sum_{i=1}^{m} \frac{1}{(\sigma^{(i)})^2}(y^{(i)} - \theta^T x^{(i)})^2$$

Compared with $J(\theta)$, it is clear that

$$w^{(i)} = \frac{1}{(\sigma^{(i)})^2}$$

(b) It is underfitting with MSE of 0.33 (2 d.p.). See Figure 6.
(c) $\tau = 0.05$ gives the minimum MSE of ≈ 0.0124 on the validation set, and gives MSE of 0.0170 on the test set. See Figures 7 to 11.

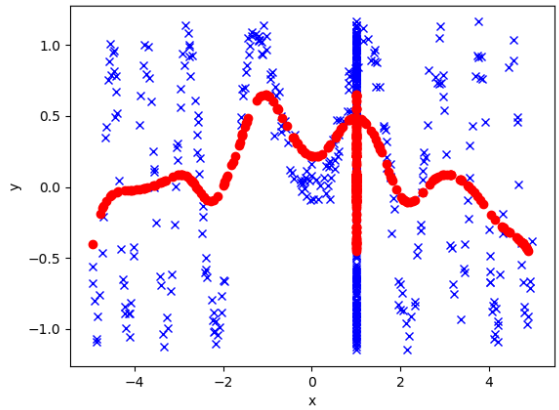

Figure 6: Prediction on the validation set with $\tau = 0.5$ compared with training set

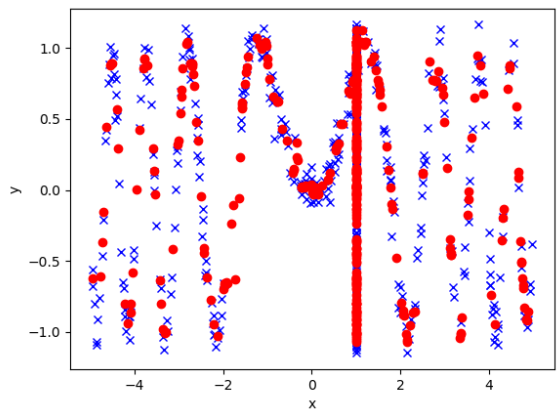

Figure 7: Prediction on the validation set with $\tau = 0.03$

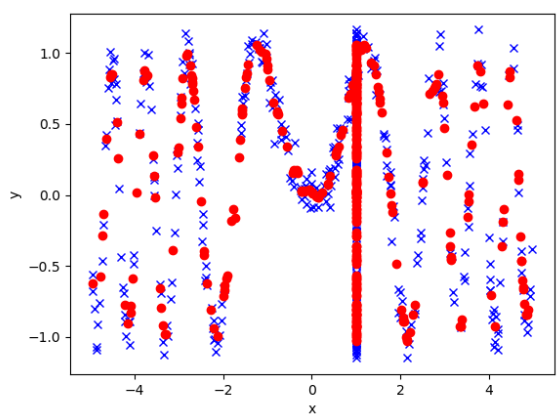

Figure 8: Prediction on the validation set with $\tau = 0.05$

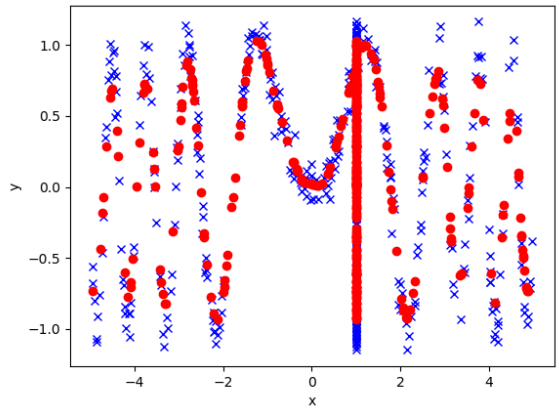

Figure 9: Prediction on the validation set with $\tau = 0.1$

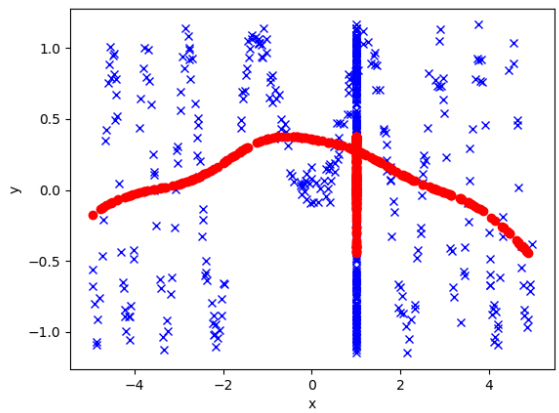

Figure 10: Prediction on the validation set with $\tau = 1.0$

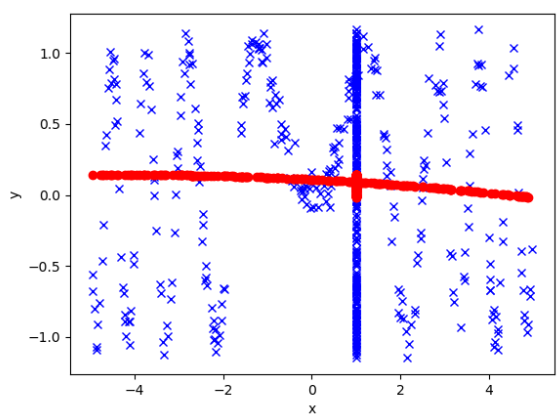

Figure 11: Prediction on the validation set with $\tau = 10.0$